

Audio Encoders and Speech Models

DATA 443/643 · Week 6, note 6.2

Week 6

American University

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Ideas from Stanford CS224S 2025 L11 (Whisper: log-mel, two convs, GPT-style decoder, 30 s, no CTC). Original slides; that course is not copied.

Lecture plan

- 1 Whisper: encoder then a causal decoder
- 2 30 seconds is 3000 frames, not 480k samples
- 3 Special tokens: language, task, time
- 4 CLAP: CLIP for audio

Key goal: pick Whisper vs CLAP by the head, not by the spectrogram.

First time: what / why / formula

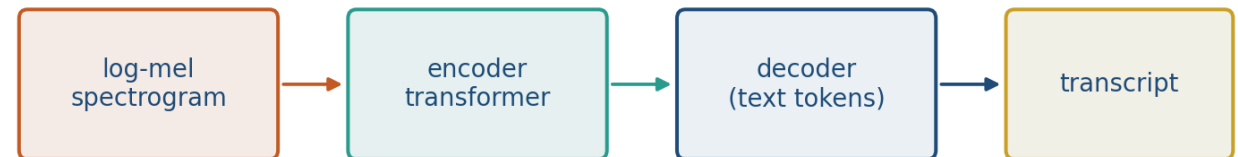
- What / why. Whisper: encoder (audio tokens) + GPT-style decoder that writes text (no CTC required in this course). CLAP: two towers like CLIP, audio instead of pixels. ASR needs a decoder. Retrieval needs a score. Do not use Whisper as a search index or CLAP as a transcriber.
- Architecture. Whisper: spectrogram $\rightarrow$ conv/embed $\rightarrow$ encoder blocks $\rightarrow$ decoder with cross-attention. CLAP: audio encoder f , text encoder g , cosine.
- Formula. Whisper LM: $\prod_t p(y_t \mid y_{<t}, \text{audio})$. CLAP: $\cos(f(a), g(t))$.
- Tradeoffs. + Whisper is a generator; CLAP is cheap retrieval. – Whisper is not a general audio tagger; CLAP does not decode words; paired audio–text is scarce (note 6.3).

How. Whisper: teacher-forced captions of speech, then generate. CLAP: InfoNCE on audio–text pairs (same loss family as 4.3).

Whisper is translation, with a spectrogram

- Log-mel in. Two convs. Transformer encoder.
- Decoder is GPT-style and causal.
CS224S: no CTC.
- Cross-attention into encoder states.
Same as note 3.3.

Whisper-style: audio encoder, text decoder (Week 3 block, new input)



Why the spectrogram exists

$$30 \text{ s} \times 16 \text{ kHz} = 480,000 \quad \text{vs} \quad 30 / 0.010 = 3000$$

3000 mel frames, 80 bands, then a stride-2 stem.

About 1500 encoder steps, not a 480k-token transformer.

Long audio: 30 s windows. Timestamp tokens shift the window.

Thirty seconds, then tokens

- Special tokens: language, `<|transcribe|>`, timestamps.
- The loss is next-token on the text, conditioned on audio.
- Timestamps are vocabulary, not a second model.

The spectrogram exists so the encoder is not a 480k-token transformer.

Raw waveform

30 s at 16 kHz
= 480,000 samples

Whisper front end (CS224S L11)

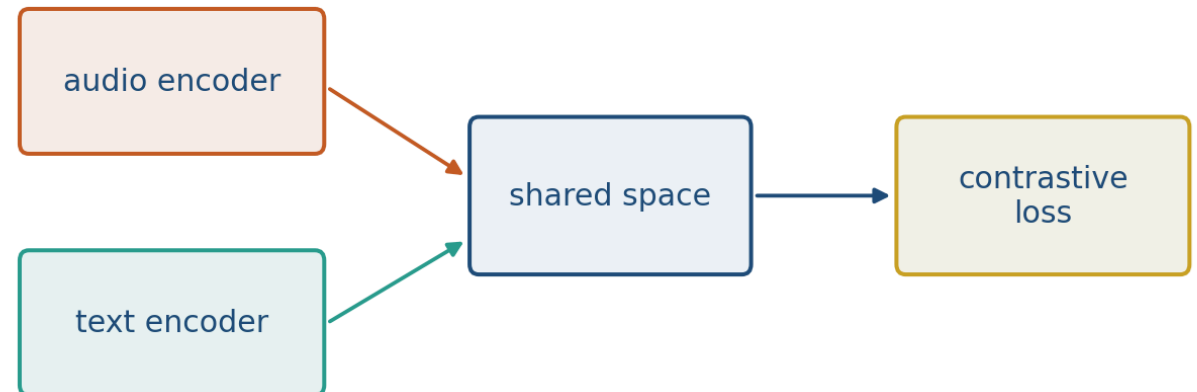
10 ms hop → 3000 mel frames

stride-2 conv → ~1500 tokens

CLAP is CLIP for sound

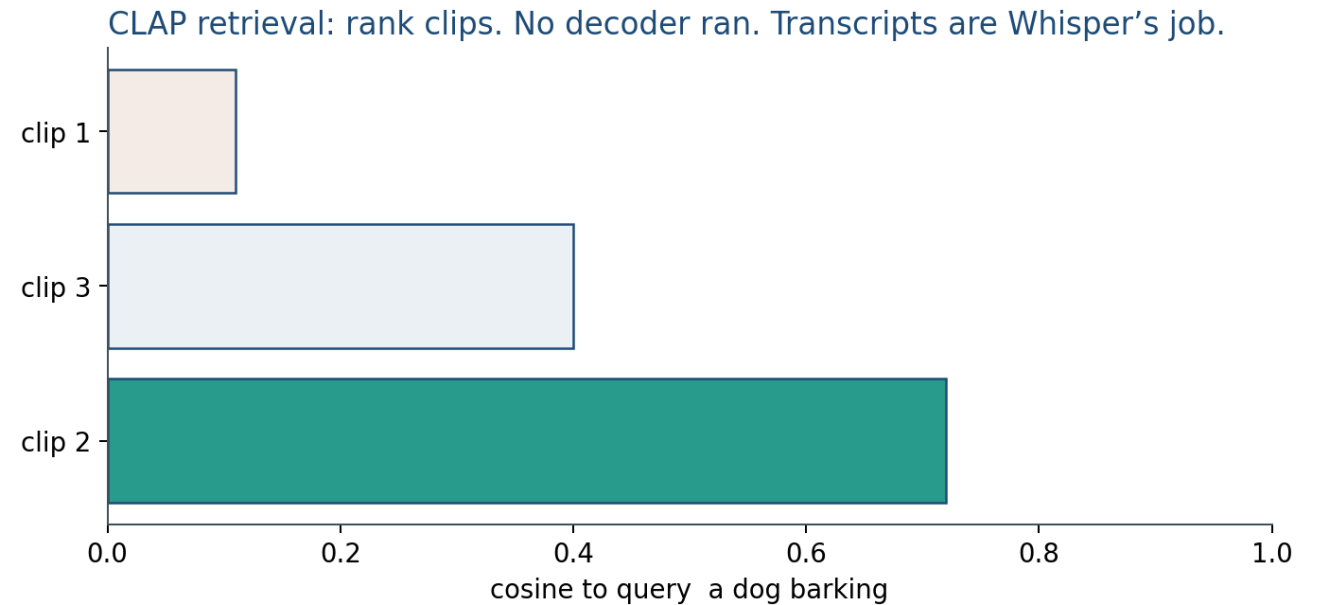
- Audio tower and text tower. InfoNCE on captions.
- Rank clips by cosine. Do not decode a transcript.
- The text tower may be bidirectional. Whisper's decoder may not.

CLAP is CLIP for audio: two towers, cosine, paired captions



Rank, then stop

- Cosines 0.11, 0.72, 0.40. Rank is clip 2, 3, 1.
- If you needed the words in clip 2, call Whisper on the wave.
- Same spectrogram tokens. Different head.



What to take from this lecture

Transcript

Whisper. Subtitles and timestamps.

Search

CLAP. Zero-shot tags. Dual retrieval.

Project

Do not concat Whisper states into CLIP without naming the rates.